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16 September 1985
SUPERSEDING
MIL-C-28892(EC)
25 March 1985

MILITARY SPECIFICATION

INTERFACE ADAPTER UNIT

This specification is approved for use by the Space and Naval Warfare Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, design, development, and test requirements for the Interface Adapter Unit (IAU), hereinafter referred to as the IAU. The IAU will interface the TSEC/KG-84, TSEC/KG-84A, or TSEC/KG-84C to shipboard secure communication systems. Hereinafter, the term KG-84 will be used to refer to all members of the KG-84 familiy, that is, the TSEC/KG-84, TSEC/KG-84A and TSEC/KG-84C. When specifying a particular member of the KG-84 family, that member is referred to by its specific name (for example, TSEC/KG-84C).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-M-13231	Marking Of Electronic Items
MIL-F-14072	Finishes For Ground Electronic Equipment
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specifications For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts); Packaging Of
MIL-C-28794	Cabinet, Electronic Equipment (Single Bay)
MIL-H-46855	Human Engineering Requirements For Military Systems, Equipment And Facilities
MIL-P-55110	Printed Wiring Boards, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5810

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MIL-STD-12	Abbreviations For Use On Drawings, And In Specifications, Standards And Technical Documents
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-188-114	Electrical Characteristics Of Digital Interface Circuits
MIL-STD-280	Definitions Of Item Levels, Item Exchangeability, Models, And Related Terms
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-740	Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810C	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1310	Shipboard Bonding, Grounding, And Other Techniques For Electromagnetic Compatibility And Safety
MIL-STD-1345	Test Requirements Document, Preparation Of
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

HANDBOOKS

MILITARY

MIL-HDBK-241	Design Guide For Electromagnetic Interference (EMI) Reduction In Power Supplies

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NATIONAL SECURITY AGENCY (NSA)

NACSIM 5100A	Compromising Emanations Laboratory Test Standard Electromagnetic
NSA 85-2	National Security Agency Performance And Interface Specification For The TSEC/KG-84C, General Purpose Telegraphy Encryption Equipment (GPEE)

JOINT TACTICAL COMMUNICATIONS OFFICE (TRI-TAC)

TT-C1-7204-0099	Performance And Interface Specification For TSEC/KG-84, Dedicated Loop Encryption Device (DLED)
TT-C1-7205-0117	Performance And Interface Specification For The TSEC/KG-84A, Dedicated General Purpose Encryption Equipment (GPEE)

NAVAL MATERIAL COMMAND (NAVMAT)

NAVMAT P4855-1	Naval Power Supply Reliability, Design And Manufacturing Guidelines

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NAVAL WEAPONS CENTER, CHINA LAKE (NAVWPNCEN)

WS-6536D Process Specification Procedures And Requirements For Preparations And Soldering Of Electrical Connections

(Copies of specifications, standards, handbook(s), and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The IAU shall provide electrical interconnection and electrical interface between two KG-84s and shipboard communications systems. The IAU design shall minimize the physical, electrical, and functional installation changes to existing secure communications systems and shall not utilize microprocessors. The IAU shall provide the waveshaping module (WSM) and level converter module (LCM) to provide proper shaping to pulse signals and to provide conversion between low and high voltage levels. Two IAUs shall be mountable as specified in 3.4.1.4. The IAU shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 IAU functional description. IAU functions shall be as specified in 3.1.2.1 through 3.1.2.4. IAU connectors are labeled and functional partitioning of the IAU is shown in FIGURE 1. FIGURE 1 is presented as a functional representation of the IAU and is not meant to be an implementation.

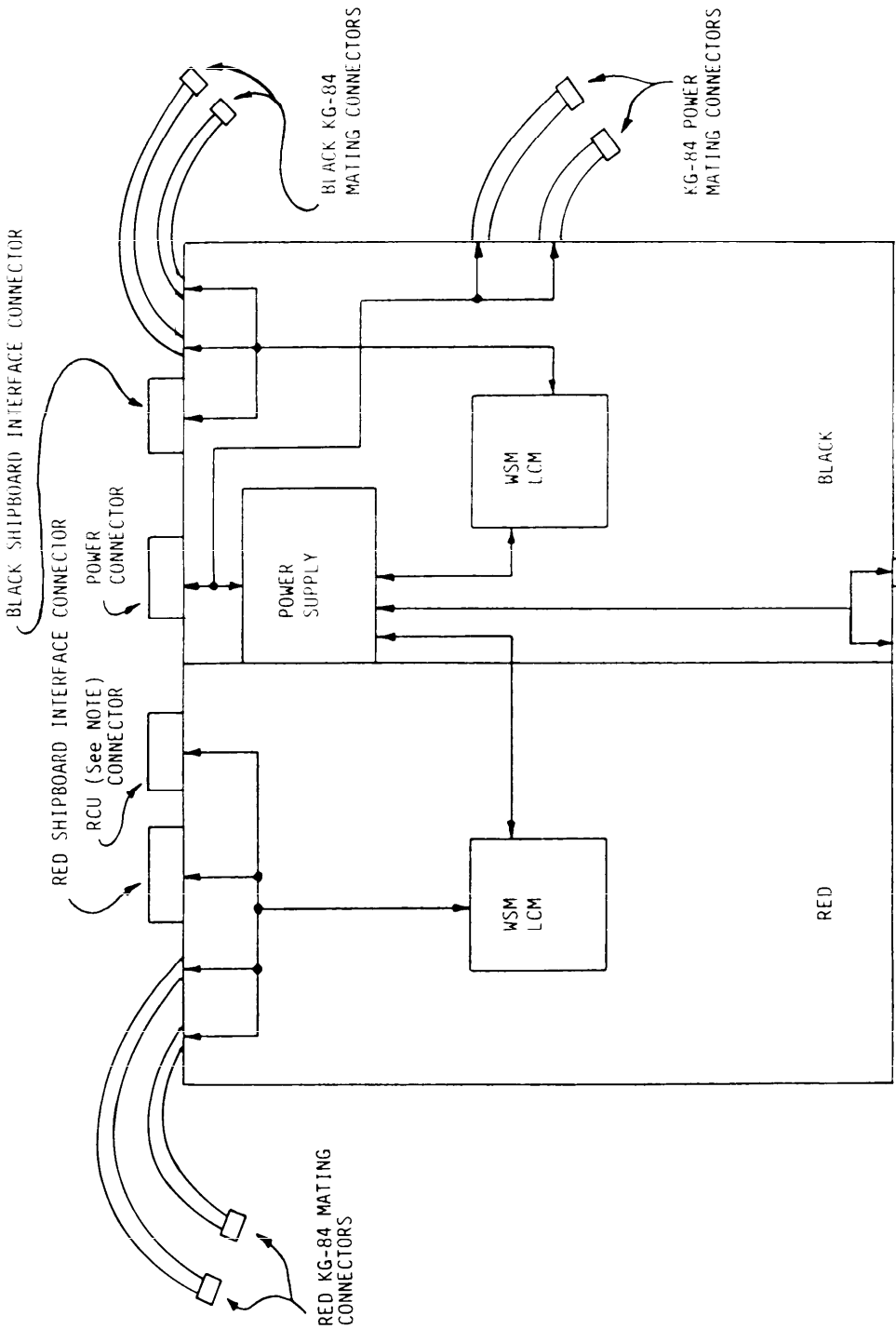
3.1.2.1 IAU General. The IAU shall provide the electrical interface for two KG-84s. The IAU shall maintain RED and BLACK signal separation. The IAU shall contain at least two WSMs and LCMs. The IAU shall connect directly to the RED and BLACK and POWER connectors of the KG-84s using shielded cabling.

3.1.2.2 WSM and LCM. The WSM and LCM shall be located on a common printed circuit board (PCB). There shall be at least one of these circuit cards in the RED section of the IAU and there shall be at least one of these circuit cards in the BLACK section of the IAU. The circuit card or cards in the RED and BLACK sections of the IAU shall be capable of interfacing to two KG-84s. The circuit card(s) utilized in the RED and BLACK sections shall be identical. There is no requirement to use the LCM when the WSM is in use and there is no requirement to use the WSM when the LCM is in use. Provisions shall be made so that either the WSM or the LCM portion of the circuit card can be used exclusive of the portion not used. When the WSM is used, signals used by the LCM and not by the WSM shall be passed through the card unmodified. When the LCM is used, signals used by the WSM and not used by the LCM shall be passed through the card unmodified. The WSM shall provide MIL-STD-188-114 waveshaping of selected RED and BLACK signals (see 3.3.2). The LCM shall provide conversion between shipboard voltage levels (normal input keying (NIK) and high level) and MIL-STD-188-114 voltage levels (low level) of the KG-84 (see 3.3.3). The LCM shall convert TXDPT and ERCT from shipboard levels (NIK and high level) to KG-84 levels (low level) and ETCT, RXDPT and STEP PULSE from KG-84 levels (low level) to shipboard levels (high level).

3.1.2.3 Strapping. In addition to other strapping requirements specified herein, provision shall be made for strapping signals specified in TABLE I, as specified in TABLE I. For all strapping options specified herein, it shall be possible to go from one option to another without the use of solder, soldering irons, or any type of wire wrapping device. All strapping options apply to both KG-84s.

3.1.2.4 KG-84 power. The IAU shall supply BLACK AC power to the KG-84s via the KG-84 power mating connectors.

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NOTE: Remote control unit

FIGURE 1. IAU functional configuration.

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TABLE I. Strapping.

Signal	Strap options
SPARE/SPARE/INVERT-P 1/	Open, +6 VDC 2/
SPARE/SPARE/RDINV-P 1/	Open, +6 VDC
SPARE/PTRR-N/UNBAL-P 1/	Open, +6 VDC, RCU connector

1/ Where differences exist in the signal name between the TSEC/KG-84, TSEC/KG-84A, and TSEC/KG-84C, the signal name for the TSEC/KG-84 will be listed first; the signal name for the TSEC/KG-84A will follow the signal name for the TSEC/KG-84; and the signal name for the TSEC/KG-84C will follow the signal name for the TSEC/KG-84A. All signal names will be separated by slashes (/). Where no differences exist in the signal name between the TSEC/KG-84, TSEC/KG-84A, and TSEC/KG-84C, only one signal name is listed.

2/ Volts direct current

3.1.3 KG-84 compatibility. The IAU shall operate with the equipments specified in a, b, and c:

a. TSEC/KG-84, in accordance with TRI-TAC TT-C1-7204-0099
b. TSEC/KG-84A, in accordance with TRI-TAC TT-C1-7205-0117
c. TSEC/KG-84C, in accordance with NSA 85-2

3.2 Interface. The IAU shall have the interfaces specified in 3.2.1 through 3.2.5. The RED input/output (I/O) interface shall be capable of accepting RED I/O signals from two KG-84s and routing signals specified in 3.2.1.1 to the IAU RED shipboard interface connector and routing signals specified in 3.2.1.2 to the IAU RCU connector for each KG-84. The BLACK I/O interface shall be capable of accepting BLACK I/O signals from two KG-84s and routing signals specified in 3.2.2 to the IAU BLACK shipboard interface connector for each KG-84. For each KG-84, RED and BLACK signals not specified in 3.2.1.1, 3.2.1.2, and 3.2.2 shall be brought into the IAU and attached to terminals (there shall be one terminal per signal) capable of accepting up to two 18 gauge wires.

3.2.1 RED I/O interface. The RED I/O interface shall be as specified in 3.2.1.1 and 3.2.1.2.

3.2.1.1 RED IAU KG-84/shipboard interface. With the exception of the signals requiring level conversion, waveshaping or strapping, the KG-84 RED signals specified in TABLE II shall be passed through the IAU between the IAU KG-84 RED mating connectors and the IAU RED shipboard interface connector unmodified. Signals which require level conversion or waveshaping shall be modified by the WSM and LCM circuit card enroute to either the KG-84 RED mating connectors or the IAU RED shipboard interface connector. General requirements for the WSM and LCM are specified in 3.1.2.2; signals which require level conversion are also specified in 3.1.2.2, and signals which require waveshaping are specified in 3.3.2. Strapping options shall be provided for these signals as specified in TABLE I and TABLE II.

3.2.1.2 RED IAU KG-84/RCU interface. With the exception of the signals requiring strapping, the KG-84 RED signals specified in TABLE III shall be passed through the IAU between the IAU KG-84 RED mating connectors and the IAU RCU connector. Strapping options shall be provided for these signals as specified in TABLE I and TABLE III.

3.2.2 BLACK I/O interface. With the exception of the signals requiring level conversion, waveshaping or strapping, the KG-84 BLACK signals specified in TABLE IV shall be passed through the IAU between the IAU KG-84 BLACK mating connectors and the IAU BLACK shipboard interface connector unmodified. Signals which require level conversion or waveshaping shall be modified by the WSM and LSM circuit card enroute to either the KG-84 BLACK mating connectors or the IAU BLACK shipboard interface connector. General requirements for the WSM and LCM are specified in 3.1.2.2; signals which require waveshaping are specified in 3.3.2. Strapping options shall be provided for these signals as specified in TABLE IV.

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TABLE II. RED IAU KG-84/shipboard interface.

SPARE/RPSTBY-N/RPSTBY-N 1/
SPARE/PTDM-P/PTCM-P 1/
SIG GND
FRAME GND
RED COMMON RET/RED COMMON RET/RXCOMB 1/
-6 TIE-OFF
RXDPT-N
RXDPT-P
PTMON-P
STEP PULSE-P
STEP PULSE-N/RED COMMON RET/TXCOMB 1/
TXCLOCK-N
TXCLOCK-P
TXDPT-N
TXDPT-P
RXCLOCK-N
RXCLOCK-P
SPARE/SPARE/RADIO-P 1/
SPARE/PTTR-P/PTTR-P 1/ 2/
RMTZER-N
SPARE/PTDM-N/RESVU-RET 1/
SPARE/PTTT-P/PTTT-P 1/
SPARE/PTTM-P/PTTM-P 1/
SPARE/PTIC-P/PTIC-P 1/
SPARE/PTTR-N/PTT, V to X 1/
SPARE/SPARE/BREAKIN-P 1/
+6 TIE-OFF
FSAO-P
FSBO-P
FSCO-P
VUCTRUP-N
RESVUCTR-N
SPARE/PTTT-N/PTTT-N 1/

1/ Where differences exist in the signal name between the TSEC/KG-84, TSEC/KG-84A, and TSEC/KG-84C the signal name for the TSEC/KG-84 will be listed first; the signal name for the TSEC/KG-84A will follow the signal name for the TSEC/KG-84; and the signal name for the TSEC/KG-84C will follow the signal name for the TSEC/KG-84A. All signal names will be separated by slashes (/). Where no differences exist in the signal name between the TSEC/KG-84, TSEC/KG-84A, and TSEC/KG-84C only one signal name is listed.

2/ Provision shall be provided for strapping this signal to +6 volts (V).

TABLE III. RED IAU KG-84/RCU.

SPARE/PTRS-N/TXRDYIND-P 1/ 2/
PTCA-P/PTRS-P/PTRS-P 1/
SXMTR-P
STEP PULSE-N/RED COMMON RET/TXCOMB 1/ 2/
TXRDY-P/PTCS-P/PTCS-P 1/
SPARE/PTCS-N/RXRDYIND-P 1/ 2/
SPARE/ALMIND-P/ALMIND-P 1/
SPARE/PTRR-P/PTRR-P 1/
SIG GND
FRAME GND
RED COMMON RET/RED COMMON RET/RXCOMB 1/
-6 TIE-OFF
+6 TIE-OFF
SPARE/SXMTR-N/CLRIND-P 1/
SPARE/PTRR-N/UNBAL-P 1/ 3/

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1/ Where differences exist in the signal name between the TSEC/KG-84, TSEC/KG-84A, and TSEC/KG-84C, the signal name for the TSEC/KG-84 will be listed first; the signal name for the TSEC/KG-84A will follow the signal name for the TSEC/KG-84; and the signal name for the TSEC/KG-84C will follow the signal name for the TSEC/KG-84A. All signal names will be separated by slashes (/). Where no differences exist in the signal name between the TSEC/KG-84, TSEC/KG-84A, and TSEC/KG-84C only one signal name will be listed.

2/ Provision shall be made to break the paths connecting these signals with the IAU KG-84 RED mating connectors and the IAU RCU connector by using straps.

3/ This signal shall only appear at the RCU connector when so strapped. Strapping for this signal is specified in TABLE I.

TABLE IV. BLACK IAU KG-84/BLACK shipboard interface.

SPARE/CTIC-N/SPARE 1/
SPARE/CTTM-N/SPARE 1/
SIG GND
FRAME GND
+6 TIE-OFF
XMT CTRL-P/CTTR-P/CTTR-P 1/
SIG COMM/SIG COMM/RXCOMB 1/
ERXC-P
ERXC-N
ETXC-P
ETXC-N
E32RXC-P/E32RXC-P/INVERT-P 1/ 2/
E32RXC-N/E32RXC-N/DP-P,BB-N 1/ 3/
E32TXC-P/E32TXC-P/UNBAL-P 1/ 2/
E32TXC-N/E32TXC-N/TRXIND-P 1/
CTCB-P/CTCS-P/CTCS-P 1/ 2/
CTCD-P/CTRS-P/CTRS-P 1/
BLK RTN/BLK RTN/TXCOMB 1/
-5 TIE-OFF
TCTC-P
TCTC-N
CLKLK-N/CTRR-P/CTRR-P 1/ 2/
CLKLK-P/CTRR-N/SPARE 1/
SPARE/CTTM-P/CTTM-P 1/
SPARE/CTIC-P/CTIC-P 1/
SPARE/TTSEL-P/TTSEL-P 1/
RMTMUX-P/RMTMUX-P/PTTO-N 1/
-5 TIE-OFF/-5 TIE-OFF/-12 TIE-OFF 1/
SPARE/CTDM-P/CTDM-P 1/ 2/
SPARE/CTDM-N/WL-N, MIL-P 1/ 3/
CTCB-N/CTCS-N/DINV-P 1/ 2/
ERCT-P
ERCT-N
ETCT-P
ETCT-N

1/ Where differences exist in the signal name between the TSEC/KG-84, TSEC/KG-84A, and TSEC/KG-84C the signal name for the TSEC/KG-84 will be listed first; the signal name for the TSEC/KG-84A will follow the signal name for the TSEC/KG-84; and the signal name for the TSEC/KG-84C will follow the signal name for the TSEC/KG-84A. All signal names will be separated by slashes (/). Where no differences exist in the signal name between the TSEC/KG-84, TSEC/KG-84A, and TSEC/KG-84C only one signal name is listed.

2/ Provision shall be made for strapping this signal to +6 VDC.

3/ Provision shall be made for strapping this signal to +6 VDC or -6 VDC.

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3.2.3 External power interface. The IAU shall not obtain its power from any of the KG-84 voltage TIE-OFF points. If the IAU requires external power, the interface shall conform to the requirements specified herein. The IAU (BLACK) interface shall provide the connection to available shipboard power. The IAU shall operate from a single-phase alternating current (AC) power source, 115 volts alternating current (VAC) ±10 percent, with a nominal frequency of 60 Hertz (Hz) ±5 percent (Type I, shipboard power as specified in DoD-STD-1399, Section 300), or 220 VAC (+10 percent, -15 percent), with a nominal frequency of 50 Hz, ±5 percent. The total power dissipated in the IAU shall not exceed 20 watts (W). Overload protection, which opens both sides of the input power lines simultaneously, shall be provided by the IAU. A chassis ground stud, as specified in MIL-STD-1310, shall be provided at the rear of the IAU. The input power shall conform to MIL-STD-454, Requirement 25.

3.2.4 Controls and indicators. If the IAU requires external power, the main power on-off switch and power indicator lamp shall be located on the IAU front panel. Placement and operation of controls and indicators shall be consistent with the Visual displays and Controls paragraphs of MIL-STD-1472.

3.2.5 KG-84 power interface. The IAU shall provide primary BLACK power to two KG-84s through the KG-84 power mating connectors. The primary power provided shall be single-phase, 115 VAC +10 percent, with a nominal frequency of 60 Hz, ±5 percent, and 220 VAC (+10 percent, -15 percent), with a nominal frequency of 50 Hz, ±5 percent. The total power required by each KG-84 will not exceed 18 W.

3.3 Performance. The IAU shall provide the modes of operation and functions specified in 3.3.1 through 3.3.3.

3.3.1 Modes of operation. The IAU shall provide for TSEC/KG-84, TSEC/KG-84A, or TSEC/KG-84C operation. The selection of TSEC/KG-84, TSEC/KG-84A, or TSEC/KG-84C need not be operator accessible when the IAU is mounted in its normal operating position. The IAU shall provide interfaces and controls for the two KG-84 devices.

3.3.2 Waveshaping. The IAU shall provide waveshaping, as specified in MIL-STD-188-114, for the ETCT, RXDPT, and STEP PULSE signals balanced or unbalanced at the 0 kilobits per second (kbps) to 2.5 kbps modulation rate and for the ETCT, TCTC, RXDPT, STEP PULSE, TXCLK, and RXCLK signals balanced and unbalanced at the 2.5 kbps to 5 kbps, 5 kbps to 10 kbps, 10 kbps to 25 kbps, and 25 kbps to 50 kbps modulation rates. The values selected for the waveshaping capacitors shall be based on the capacitance at the KG-84 output drivers. It shall be possible to disable the waveshaping of any of these signals independent of the others.

3.3.3 Level conversion. The IAU shall provide level conversion (high-to-low, NIK-to-low, and low-to-high) for the signals specified in 3.1.2.2. Using straps, it shall be possible to choose whether TXDPT and ERCT are converted from high level to low level or NIK to low level. Specifications for the high-to-low converter are provided in TABLE V. Specifications for the NIK-to-low converter are provided in TABLE VI, and specifications for the low-to-high converter are provided in TABLE VII.

3.4 Characteristics. The physical, environmental, and life cycle characteristics of the IAU shall be as specified in 3.4.1 through 3.4.2.6.

3.4.1 Physical characteristics. The physical characteristics of the IAU shall be as specified in 3.4.1.1 through 3.4.1.4. The IAU, as delivered, shall include all mating connectors and backshells for the mating connectors.

3.4.1.1 Weight. The total weight of two IAUs, including the mount and cables, shall not exceed 40.5 kilograms (kg) (90 pounds (lbs)).

3.4.1.2 Dimensions. The height, width, and depth of the IAU shall not exceed 12.7 centimeters (cm) (5.0 inches (in.)) by 44.5 cm (17.5 in.) by 45.7 cm (18.0 in.), respectively.

3.4.1.3 Transport and storage requirements. The IAU shall be designed for transportation in any form of military or commercial common carrier. The use of a carrying case to provide protection shall be subject to approval by the procuring activity.

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3.4.1.4 Mounting provisions. Two IAUs shall be mountable on a tray in a 19 in. rack conforming to MIL-C-28794. The tray when loaded with two IAUs shall be mountable within a 26.67 cm (10.5 in.) vertical space in the rack. FIGURE 2 illustrates the location of mounting holes in the rack. Referring to FIGURE 2, the tray shall be mountable using triangular bolt hole pattern sets such as T1 and T2 on the left hand rack side wall and the matching set of triangular bolt hole patterns found on the right hand rack side wall. The lowest edge of the mounted IAU pair and tray shall be not lower than 2.87 cm (1.13 in.) below the center of the hole designated A in FIGURE 2 and the highest edge of the mounted IAU pair and tray shall be not higher than 2.87 cm (1.13 in.) below the center of the hole labeled B in FIGURE 2.

TABLE V. High level to low level converter.

Data rates of operation	DC 1/ TO 300 bps 2/
Input	The input shall be capable of operating with a nominal 10 mA 3/ to 60 mA unipolar current
	The input shall be reverse polarity protected
	The maximum voltage drop across the input shall not exceed 4 VDC with a 100 mA input current
Input threshold	Shall be 10 mA
Output signal	Shall be MIL-STD-188-114. Shall be selectable for either balanced or unbalanced
I/O phase relationship	The phase relationship of the I/O signals shall be selectable as either in-phase or inverted
DC I/O isolation	Shall be greater than 100 megohms between input and output
Breakdown voltage	Shall be 500 V input to output
AC coupling	Less than 25 pF 4/ input to output

1/ Direct current
2/ Bits per second
3/ Milliampere
4/ Picofarads

TABLE VI. NIK to low level converter.

Data rates of operation	DC to 300 bps
Input	The input of the NIK converter shall provide a 2 VDC (±0.5 VDC) 70 µA 1/ source which is externally switched. In the on condition the resistance of the external switch does not exceed 300 ohms. In the off state the resistance of the external switch is greater than 100 kilohms
	With the external switch in the on state the current through the switch shall not exceed 80 µA
	With the external switch in the off state the voltage across the switch shall not exceed 2 VDC (±0.5 VDC)
Input threshold	Shall be 60 µA
Output signal	Shall be MIL-STD-188-114. Shall be selectable for either balanced or unbalanced
	Shall be short circuit protected

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TABLE VI. NIK to low level converter. (continued)

I/O phase relationship	The phase relationship of the I/O signals shall be selectable as either in-phase or inverted
DC I/O isloation	Shall be greater than 100 megohms between I/O
Breakdown voltage	Shall be 400 V input to output
AC coupling	Less than 25 pF input to output

1/ Microampere

TABLE VII. Low level to high level converter.

Data rates of operation	DC to 300 bps
Input signal	MIL-STD-188-114 (±25 VDC maximum) balanced or unbalanced
Input impedance	Greater than 4000 ohms
Input threshold	±500 millivolts
Output	The output shall be capable of switching an external 20 mA to 60 mA teletype signal
	The voltage drop across the output when it is in the low resistance or on state shall not exceed 4 VDC at 100 mA
	When the output is in the high resistance or off state the leakage current through the output shall not exceed 100 mA with 200 VDC across the output
I/O phase relationship	The phase relationship of the I/O signals shall be selectable as either in-phase or inverted
DC I/O isolation	Shall be greater than 100 megohms between I/O
Breakdown voltage	500 VDC input to output

3.4.2 Environmental requirements. The IAU shall be designed and constructed to resist and withstand the Range 4 environmental conditions specified in MIL-E-16400 and in 3.4.2.1 through 3.4.2.6. Conformance to the requirements specified in 3.1 through 3.4.1.4 shall be achieved before, during, and after exposure to the environmental conditions.

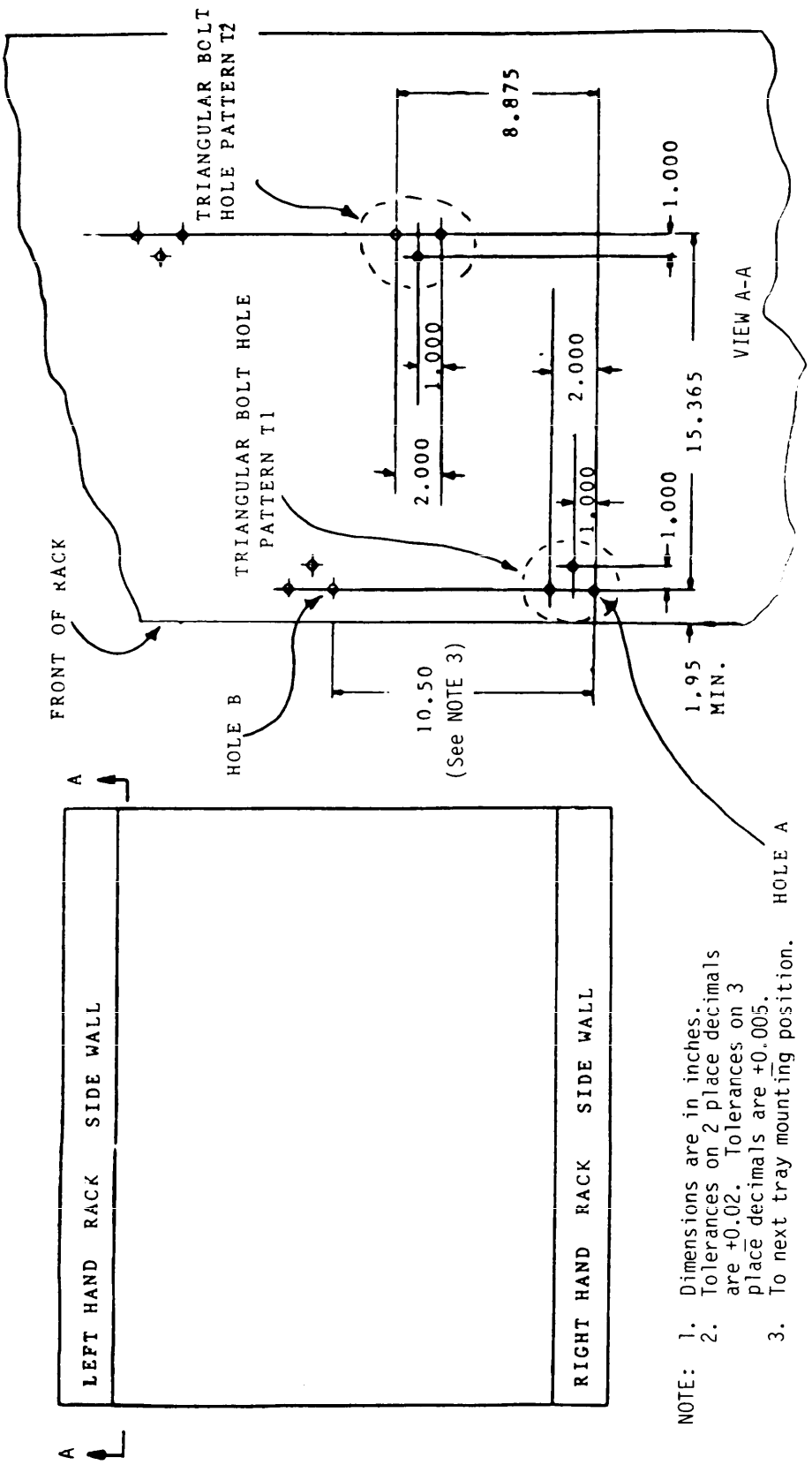
3.4.2.1 Temperature extremes. The IAU shall be designed to withstand the temperature conditions specified in 3.4.2.1.1 and 3.4.2.1.2.

3.4.2.1.1 Storage and transit. The IAU, in the nonoperating mode, shall withstand the inclusive extremes of -62° Celsius (C) to +71°C ambient air temperature.

3.4.2.1.2 Continuous operating temperature range. The IAU shall function without degradation of performance when operated continuously within the inclusive ambient air temperature range of 0°C to +50°C.

3.4.2.2 Humidity. The IAU shall sustain no physical damage or degradation of performance when exposed in the operating mode to a relative humidity of 95 percent for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the equipment in the form of both water and frost.

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NOTE: 1. Dimensions are in inches.
2. Tolerances on 2 place decimals are ±0.02. Tolerances on 3 place decimals are ±0.005.
3. To next tray mounting position.

FIGURE 2. Hole patterns.

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3.4.2.3 Fungus. The IAU shall be resistant to fungi and shall not be adversely affected by fungi under conditions favorable to fungus growth (high humidity, warm atmosphere, and presence of inorganic salts).

3.4.2.4 Salt fog. The IAU shall withstand prolonged exposure in a salt-laden atmosphere without degradation of materials, finishes, or performance, and shall comply with the salt fog requirements of MIL-STD-810C, Method 509.2, Procedure I.

3.4.2.5 Shock. The IAU shall comply with the Shock paragraph of MIL-E-16400, and shall withstand the shock requirements of MIL-S-901 for Grade A, Class I, Type A.

3.4.2.6 Vibration. The IAU shall withstand the Type I vibration test of MIL-STD-167-1.

3.5 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials, and processes paragraph of MIL-E-16400, except as specified in 3.5.1 through 3.5.10.

3.5.1 Parts control. Electrical and mechanical parts to be incorporated in the IAU shall be controlled in accordance with MIL-STD-965, Procedure I. Nonstandard parts for power supplies shall be available from at least two sources.

3.5.2 Connectors. Connectors shall conform to the Connectors, electrical and radio frequency and the Mechanical design and construction paragraphs of MIL-E-16400, except as specified in TABLE VIII. KG-84 mating connector backshells shall conform to the applicable MS part No. or Glenair part No., or equal, specified in TABLE VIII. KG-84 mating connectors shall conform to the applicable MS part No. specified in TABLE VIII. The RED shipboard interface connector, RCU connector, and BLACK shipboard interface connector shall each have a minimum of 10 percent spare pins. Each spare pin on the RED shipboard connector, RCU connector, and BLACK shipboard interface connector shall be connected to a terminal capable of accepting up to two 18 gauge wires (one terminal per spare pin). The IAU shall utilize one RED shipboard interface connector, one BLACK shipboard interface connector, and one RCU connector.

TABLE VIII. Backshell connectors and mating connectors.

Backshell connector	MS part No.	Glenair part No., or equal	Mating connectors MS part No.
RED	MIL-M-85049/3616-W05	380FS001B1608A3	MS27467T17B35S
BLACK	MIL-M-85049/3616-W05	380FS001B1608A3	MS27467T17B35SA
Power	MIL-M-85049/3614-W04	380FS001B1606A3	MS27467T15B18SA

3.5.3 Cables. Shielded cables shall be provided to connect the IAU with the KG-84s. The cables shall be long enough to allow maintenance personnel access to the back of the KG-84s for removal of the equipment. The cables shall terminate with the connectors specified in TABLE VIII.

3.5.4 Protective finish. The IAU shall be finished in accordance with MIL-F-14072. The final paint film on exterior surfaces shall be semigloss light-gray enamel, Color Number 26152 of FED-STD-595.

3.5.5 Conformal coating. Printed wiring assemblies shall be conformally coated in accordance with MIL-STD-454, Requirement 17. Adjustable portions of components and mating surfaces (such as connector contact points, test points, and screw threads) shall not be coated.

3.5.6 Batteries. The IAU shall not utilize batteries.

3.5.7 Power supply. If required, the power supply shall conform to the requirements specified in 3.5.7.1 through 3.5.7.3.

3.5.7.1 Power density. The power density shall be output power divided by power supply envelope volume, including electromagnetic interference (EMI) filtering where required. Power density exceeding 2 W per cubic in. shall be approved by the procuring activity.

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3.5.7.2 Temperature limits. Except as otherwise specified herein, junction temperature of semiconductor devices (diodes, transistors, hybrid, and monolithic integrated circuits, and so forth) shall not exceed +110°C under worst-case conditions. Hot spots on the case shall not exceed the temperature specified in a through d:

a. 40°C rise above external device ambient with a maximum absolute temperature of +110°C for devices less than or equal to 3 W of dissipation
b. 55°C rise above external device ambient with a maximum absolute temperature of +125°C for devices greater than 3 W of dissipation
c. 30°C rise above external device ambient with a maximum absolute temperature of +100°C for transformers
d. 10°C rise above external device ambient, due to self-heating, with an absolute maximum temperature of 85°C for capacitors.

3.5.7.3 Design and manufacture. The power supply shall be designed to be maintainable at either intermediate or depot level; it shall not be limited by design to repair by the manufacturer. The power supply shall not be encapsulated or embedded (potted) unless shown by analyses and test to be necessary for heat removal or dissipation. This requirement shall exclude conformal coating. The power supply may be tested with the end equipment, or may be tested as a unit by duplicating the input provided to the power supply as installed for operation in the end equipment. EMI requirements shall be tailored for the power supply which will enable the end equipments to conform to the required Class of MIL-STD-461. Guidance for design and manufacturing of power supplies is contained in NAVMAT P4855-1.

3.5.8 Printed wiring assemblies. Materials and processes for printed wiring assemblies shall conform to MIL-P-55110.

3.5.9 Soldering. Soldering shall conform to the requirements of WS-6536D.

3.5.10 Parts rescreening. All hybrids, semiconductors, and integrated circuits procured for the IAU (including IAU spares) shall be subject to parts rescreening at incoming inspection.

3.6 Design and construction. The electrical and mechanical design of the IAU shall be in accordance with the Electrical design and construction and Mechanical design and construction paragraphs of MIL-E-16400, except that the paragraphs specified in a through d do not apply:

a. Dielectric withstanding voltage
b. Insulation resistance
c. Synchro data transmission systems
d. Soldered connections

3.6.1 TEMPEST. The IAU shall comply with the TEMPEST requirements specified in NACSIM 5100A.

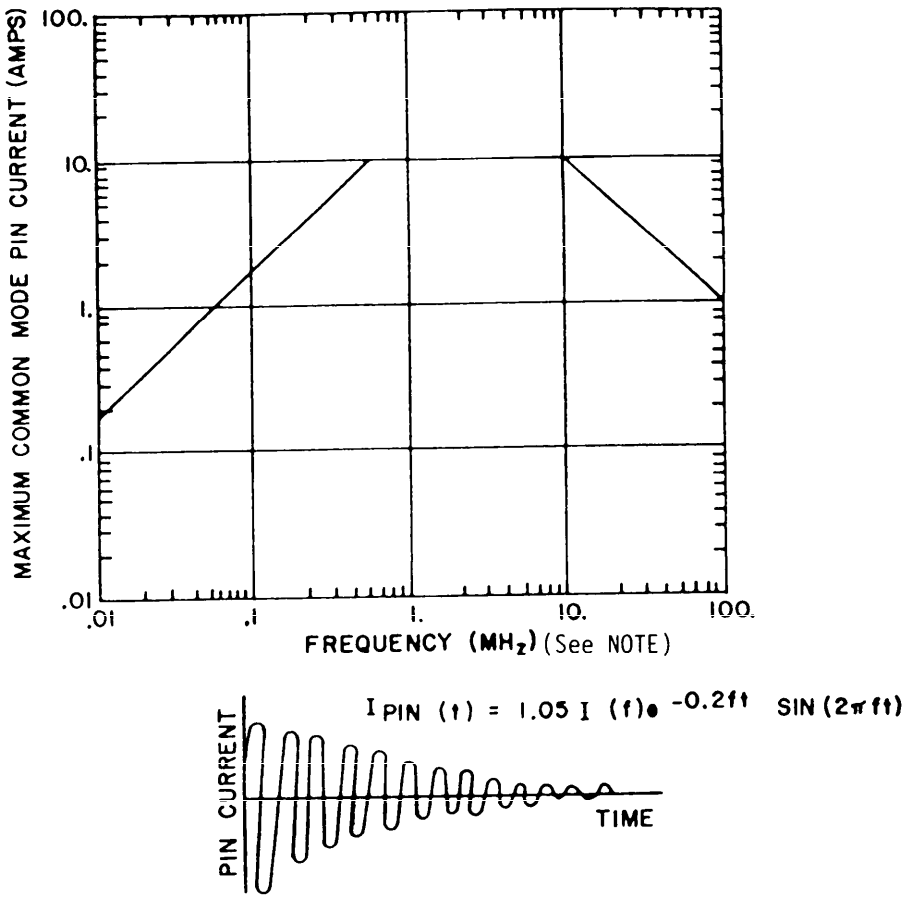
3.6.2 EMI. Except as specified in 3.6.3, the IAU shall comply with the requirements of MIL-STD-461 for Class A4 equipments. Guidance for limiting EMI is provided in MIL-HDBK-241.

3.6.3 Electromagnetic pulse (EMP). The IAU shall be hardened or shall be intrinsically hard against the EMP resulting from an exo-atmospheric nuclear burst to survive the levels specified herein. All equipment and subsystem interface pins and terminals of leads, including grounds and neutrals, shall not exhibit permanent malfunction and shall resume normal operation within 10 seconds after being subjected to a test signal having either the waveform and common mode current level shown on FIGURE 3 or the waveform and common mode voltage level shown on FIGURE 4, whichever occurs first, at the specific test frequency.

3.6.4 Transient protection. The IAU shall conform to the requirements for transients as specified in DoD-STD-1399, Section 300. The IAU shall survive undamaged, but need not operate through the 2500-V spike and emergency conditions. The IAU shall continue normal operation, with no degradation of performance, when subjected to 1000-V spikes and the other voltage and frequency variations and transients specified.

3.6.5 Degree of enclosure. The degree of enclosure of the equipment shall be dripproof (15 degrees) in accordance with MIL-STD-108.

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NOTE: Megahertz

FIGURE 3. Limit for common mode current.

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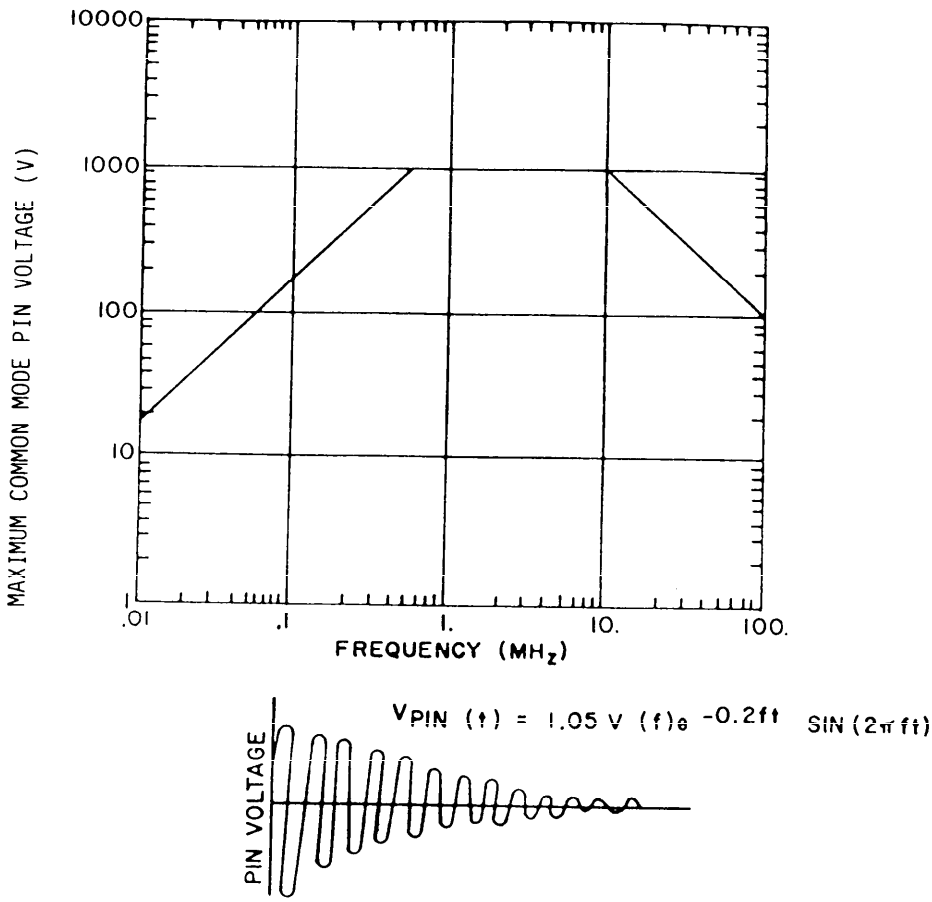


FIGURE 4. Limit for common mode voltage.

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3.6.6 Human performance. The IAU shall comply with the human factors, operational, and maintenance design criteria of MIL-STD-1472 and the requirements specified in 3.6.6.1 through 3.6.6.3.

3.6.6.1 Design for accessibility. The design of the IAU shall be in accordance with the requirements of the Accessibility paragraph in MIL-E-16400, and MIL-STD-1472, except as specified in 3.6.6.2. IAU circuit cards and power supply shall be accessible from the front of the unit.

3.6.6.2 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintainer panels, shall conform to MIL-STD-1472 and shall be in accordance with the Controls, indicators and panel layout paragraph of MIL-E-16400, except that color coding of lamps and indicators shall be in accordance with MIL-STD-1472. Color coding and marking of pin jacks and other test points shall be in accordance with MIL-STD-415. The abbreviations of MIL-STD-12 shall be used for equipment labeling, where applicable.

3.6.6.3 Noise constraints. The IAU shall conform to the Type 3 structure-borne noise constraints specified in MIL-STD-740.

3.6.7 Safety criteria. The IAU design shall conform to the Safety criteria paragraph of MIL-E-16400, except as specified in 3.6.7.1 through 3.6.7.5.

3.6.7.1 Unacceptable materials. IAU design shall not include the materials specified in a through g:

a. Polychlorinated biphenyls
b. Asbestos and asbestos compounds
c. Flammable or combustible materials
d. Fragile or brittle materials
e. Beryllium and beryllium compounds unless so identified
f. Lithium and lithium compounds unless approved by the procuring activity
g. Mercury or mercury-containing components unless approved by the procuring activity

The Unacceptable materials paragraph of MIL-E-16400 shall apply.

3.6.7.2 Printed circuit assembly protection. With power applied to the IAU, removal or insertion of printed circuit assemblies shall not damage the assembly or any other part of the IAU. Each assembly shall be keyed to prevent incorrect insertion.

3.6.7.3 Electrical cable protection. The IAU design shall preclude damage to electrical cabling during all conditions of assembly, removal, insertion, and tilting of the IAU.

3.6.7.4 Leakage current. The leakage current of the IAU shall not exceed 5 mA to ground.

3.6.7.5 DC reverse polarity protections. For each DC input, the IAU shall contain reverse polarity protection.

3.6.8 Interchangeability. Interchangeability is required at all item levels specified in MIL-STD-280. An interchangeable item possesses functional and physical characteristics equivalent in performance, safety, reliability, and maintainability, to another item of similar or identical purposes, and is capable of being exchanged for the other item without any adjustment or alteration of the items, or of adjoining items. Interchangeability shall be demonstrated at the major parts, subassembly, and equipment item levels.

3.7 Software design. The design of the IAU shall not utilize microprocessors or software.

3.8 Identification and marking. The nomenclature, identification, marking, and labels of the IAU shall be in accordance with MIL-M-13231. Clean, concise, and durable markings shall be provided for all controls, lamps, switches, fuses, jacks, test points, and other components.

3.9 Life-cycle characteristics. The life cycle characteristics shall be as specified in 3.9.1 through 3.9.3.3.

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3.9.1 Reliability. The IAU shall be designed to conform to a predicted upper test mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) of 5000 hours and shall have a lower-test MTBF (θ_1 as defined in MIL-STD-781) of 2500 hours. Power supply MTBF shall be not less than 40,000 hours at 55°C with 110°C maximum junction temperature.

3.9.2 Maintainability. The IAU shall have a mean-time-to-repair (MTTR) of 15 minutes and a maximum corrective maintenance time (M_{maxct}) at the 95th percentile of 45 minutes when repair is accomplished at the organizational level by the replacement of modules, PCBs, and chassis-mounted parts and components. Corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance tasks. The IAU shall have the test and maintenance characteristics specified in 3.9.2.1 through 3.9.2.3.

3.9.2.1 Built-in-test (BIT). The IAU shall not utilize BIT.

3.9.2.2 General purpose electronic test equipment (GPETE). GPETE shall conform to MIL-STD-1364. The Test equipment and tool list paragraph of MIL-STD-1345 is applicable for non-standard GPETE and special purpose electronic test equipment.

3.9.2.3 Off-line automatic test equipment (ATE) provision. The IAU shall not utilize ATE.

3.9.3 Maintenance. It shall be possible to gain access to IAU circuit cards and power supply from the front panel of the unit. IAU maintenance shall be as specified in 3.9.3.1 through 3.9.3.3.

3.9.3.1 Maintenance concept. Two maintenance levels shall be applied to the equipment: organizational and depot.

3.9.3.1.1 Organizational maintenance. The IAU shall be designed to maximize the ability of organizational maintenance personnel to restore the IAU to full operational capability by fault isolation to the lowest replaceable unit (LRU) and to replace that LRU from authorized spares. LRU in this context is a replaceable subassembly (such as a slide-in printed circuit assembly) or chassis-mounted part (such as front panel switches).

3.9.3.1.2 Depot maintenance. The depot shall repair LRUs that are beyond the capabilities of the organizational maintenance level. Depot level repair shall allow testing to provide fault isolation to piece parts on the replaceable subassemblies separated from the IAU and replacement of those piece parts.

3.9.3.2 Preventive maintenance. The IAU shall not require regularly scheduled preventive maintenance, such as alignment and adjustment. The IAU shall be capable of operating for at least 90 days without preventive maintenance.

3.9.3.3 Special tools and test equipment. Special tools (see 6.4) and test equipment shall not be required to maintain the IAU.

3.10 Workmanship. Workmanship shall be in accordance with 3.10.1 and 3.10.2.

3.10.1 General workmanship. Workmanship shall conform to the Workmanship paragraphs of MIL-E-16400.

3.10.2 Workmanship screen. The IAU shall withstand a defect detection vibration screen of random type vibration (see 4.8.2.1) at 0.04 g^2/Hz ±3 decibels (dB) from 80 Hz to 350 Hz (and 3 dB/octave fall-off outside these limits) and temperature cycling (see 4.8.2.2) with operation at the maximum and minimum operating temperatures and dwell time at the nonoperating temperature extremes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
    3. Environmental inspection (Group C) (see 4.4.3)
c. Safety verification (see 4.5.12)
d. Inspection of preparation for delivery (see 4.9)

4.3 First article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include all tests and examinations specified in TABLE IX. The number of units to undergo each first article examination or test shall be as specified in TABLE IX and 6.2.1. First article inspection shall be performed in an order acceptable to the Government; however, Group A inspections, which shall be performed on each first article unit, shall be performed prior to Group B and Group C inspections.

TABLE IX. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
			Number of units	Group A	Group B	Group C
Weight	3.4.1.1	4.5.1a	10	X		
Dimensions	3.4.1.2	4.5.1b	10	X		
Parts, materials, and processes	3.5	4.5.1c	10	X		
Identification and marking	3.8	4.5.1d	10	X		
IAU power	3.2.3	4.5.3.1	10	X		

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TABLE IX. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
			Number of units	Group A	Group B	Group C
EMI	3.6.2	4.5.8	2			C-II
TEMPEST	3.6.1	4.5.7	2			C-II
External power	3.2.3, 3.6.4	4.5.6	2			C-I
Continuity and isolation	3.1, 3.2	4.5.5.1	2		X	
Waveshaping	3.3.2	4.5.5.2	2		X	
Level conversion	3.1.2.2, 3.3.3	4.5.5.3	2		X	
Temperature	3.4.2.1	4.5.4a	1			C-I
Humidity	3.4.2.2	4.5.4b	1			C-I
Salt fog	3.4.2.4	4.5.4c	1			C-III
Fungus	3.4.2.3	4.5.4d	1			C-III
Shock	3.4.2.5	4.5.4e	1			C-III
Vibration	3.4.2.6	4.5.4f	1			C-II
EMP	3.6.3	4.5.9	1			
Enclosure	3.6.5	4.5.10	1			C-I
Human engineering	3.6.6	4.5.11	1			
Maintainability	3.9.2	4.7	1			
Reliability	3.9.1	4.6	12			
Safety	3.6.7	4.5.12	2			
Interchange-ability	3.6.8	4.7.1	1			
General workmanship	3.10.1	4.8.1	10	X		
Workmanship screen	3.10.2	4.8.2	10	X		

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be performed on every IAU offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal the omissions and errors of the production process such as limited functional and performance tests, tests which detect deviations from design, and tests of controls and adjustments. Production inspection shall include the examinations and tests specified in Group A of TABLE IX. The workmanship screen shall be the first test performed for each IAU.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. Production control inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts. The tests shall be performed on the complete IAU as offered for delivery. Production control inspection shall include the examinations and tests specified in Group B of TABLE IX.

4.4.2.1 Sampling plan for production control inspection (Group B). IAUs submitted for production control inspection shall be selected by the contractor under the supervision of the Government quality assurance representative (QAR) and shall be representative of current production. Sampling shall conform to the procedures for small sample inspection of MIL-STD-105, using the special inspection levels. Production control inspection shall be performed on inspection lots that have passed Group A inspection. The acceptable quality level shall be a 6.5 percent and the inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed in an order acceptable to the Government.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective IAUs and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

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4.4.3 Environmental inspection (Group C). Environmental inspection shall be performed on a sampling basis as specified in 4.4.3.1. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the IAU as a whole; tests of the effects of changes in the environment (such as extremes of temperature and humidity); tests of changes to the environment due to the IAU (such as TEMPEST); and tests of the effects of shock and vibration. Environmental inspection shall include the examinations and tests specified in Group C of TABLE IX. Group C inspection shall be performed in an order acceptable to the Government.

4.4.3.1 Sampling plan for environmental inspection (Group C). Two units shall be selected from every other Group B sample. All of the Group C tests shall be distributed among the two units at the frequency specified herein; however, the assignment of specific tests to each unit shall be at the discretion of the contractor. The pairs of selected units shall be subjected to testing as specified in a through c:

a. Subgroup C-I: first, fourth, seventh Group C sample (every third Group C sample)
b. Subgroup C-II: first, seventh, thirteenth Group C sample (every sixth Group C sample)
c. Subgroup C-III: first, tenth, nineteenth Group C sample (every ninth Group C sample)

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the IAU to conform to the specified requirements, or if the contractor cannot determine the cause of the failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming production control and environmental sample units. Unless otherwise specified (see 6.2.1), sample units which have been subjected to, and have passed, production control and environmental inspections may be accepted on the contract provided all damage is repaired and the sample units are resubjected to, and pass, production inspection.

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.12.3.

4.5.1 Surface examination. The IAU shall be examined to determine conformance to the requirements specified in a through f:

a. Weight (see 3.4.1.1)
b. Dimensions (see 3.4.1.2)
c. Parts, materials, and processes (see 3.5)
d. Identification and marking (see 3.8)
e. Finish (see 3.5.4)
f. Mounting provisions (see 3.4.1.4)

Particular attention shall be given, but not limited, to the examinations specified in g through r:

g. Completeness
h. Operation of adjustable and sliding parts, thumb screws, controls, and switches
i. Welded and soldered joints
j. Fit of components in their respective positions
k. Check of mounting means
l. Lubrication and rust prevention
m. Safety measures and interlocks
n. Fastening and securing devices or parts
o. Cable runs between components, including plugs and receptacles
p. Grounding connections
q. Foreign objects, lint, and dust
r. Potential shorts and defective insulation

In the examination, the IAU shall not be disassembled in a manner that will affect the performance, durability, or appearance of the IAU.

4.5.2 Parts rescreening. Rescreening tests shall be performed on hybrids, semiconductors, and integrated circuits at incoming inspection. The rescreening shall include functional performance testing at -25°C, +25°C, and +125°C.

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4.5.3 Performance tests. The IAU shall be tested to determine conformance to the operation and performance requirements specified in 3.3. The test specified in 4.5.3.1 shall be performed.

4.5.3.1 IAU power. If external power is required, verify that the ON lamp is lit when power is applied.

4.5.4 Environmental tests. The IAU shall be subjected to the tests of MIL-E-16400 specified in a through f. The IAU shall be mounted in the tray specified in 3.4.1.4 during the tests.

a. Temperature (Range 4)
b. Humidity
c. Salt fog
d. Fungus (required only for nutrient materials other than those of MIL-STD-454, Requirement 4, the Materials (for new designs) paragraph)
e. Shock (MIL-S-901, Type A, Grade A, Class I for shipboard)
f. Vibration (MIL-STD-167-1, Type I for shipboard)

4.5.5 Electrical performance tests. Tests specified in 4.5.5.1 through 4.5.5.3 shall be performed to verify that the IAU conforms to the requirements for specific signals and tolerances specified in 3.1, 3.2, and 3.3.

4.5.5.1 Continuity and isolation. The continuity of every signal, from the KG-84 interface to the external interface shall be checked. Every pin not associated with a given signal shall be checked to verify isolation of the selected signal from all other signals. Operation of strap options shall also be verified.

4.5.5.2 Waveshaping. The rise and fall time of each of the specified signals shall be measured for each specified data rate to verify conformance to the waveshaping requirements of MIL-STD-188-114.

4.5.5.3 Level converters. For each level converter (NIK-to-low, high-to-low, and low-to-high) verify conformance to the characteristics specified in TABLE V, TABLE VI, and TABLE VII.

4.5.6 External power. The IAU shall be subjected to the tests specified in the Electrical tests paragraph of MIL-E-16400, except as specified by 3.6.4.

4.5.7 TEMPEST. The IAU shall be subjected to TEMPEST tests in accordance with a test plan approved by the procuring activity.

4.5.8 EMI test. The IAU shall be subjected to EMI tests in accordance with MIL-STD-462 and a test plan approved by the procuring activity.

4.5.9 EMP. The EMP requirements of 3.6.3 shall be verified by analysis or test of a pulse induced from a simulated exo-atmospheric nuclear explosion whose radio-frequency (RF) field is of the form and amplitude specified herein. The KG-84 shall not be connected to the IAU during the test. Verification by testing shall be accomplished as specified in a through f:

a. Applicable test frequencies. Tests shall be performed for the injection frequencies of 0.01 MHz, 0.1 MHz, 0.5 MHz, 1 MHz, 2 MHz, 5 MHz, 10 MHz, 20 MHz, 50 MHz, and 100 MHz. The test shall be performed at the test sample's critical frequencies, such as local oscillator, power switching frequencies and harmonics, RF, clock frequencies, bandpass frequencies, and so forth, as specified in the EMI test plan.

b. Apparatus. Test apparatus shall consist of the items specified in 1 through 4:

1. Source injection pulse generator with the capabilities specified in A through D:

A. Damped sinusoidal transient as shown in the applicable limit and as described by the equations specified in (1) and (2):

(1) $I_{pin}(t) = 1.05I(f)e^{-0.2ft}\sin(2\pi ft)$

(2) $V_{pin}(t) = 1.05V(f)e^{-0.2ft}\sin(2\pi ft)$

$I_{pin}(t)$ = Pin to case current, in amperes at time, t

$V_{pin}(t)$ = Pin to case voltage, in V at time, t

f = Specific test frequency, in MHz

t = Time in seconds

I(f) = Current determined from applicable limit curve of FIGURE 3

V(f) = Voltage determined from applicable limit curve of FIGURE 4

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B. Variable amplitude so the injected transient can be increased from a maximum specified level (that is, at least 10 percent; 50 percent of the amplitude shall be tested at a specified test frequency).
C. Frequency range capable of generating the required test frequencies
D. Current output level of at least 10 amperes into a 100-ohm load

2. Current probe
3. Oscilloscope with a minimum bandwidth capability of 100 MHz
4. Interconnecting test cable

c. Test setup and procedure. The test setup and procedure shall be as specified herein. A functional analysis shall be performed before the certification testing to identify the items specified in 1 through 3:
1. The interface pin connectors
2. The test sample functions which shall be monitored to determine system failure or transient upset
3. The additional test frequencies (that is, interface cable resonant frequency, clock frequencies, data rates, and so forth)

d. Test setup. Test setup shall be as shown in FIGURE 5.
e. Test procedure. The test procedure shall be as specified in 1 through 5:
1. Apply the test signal to each interface of the test sample, at each test frequency.
2. Monitor the test sample functions in all modes of operation to determine whether there is a failure or transient upset
3. Inject one transient pulse on each interface pin at each test frequency.
4. The test sample shall be tested with the power off and on.
5. Record test frequency and level at which a failure or upset occurs

f. Test sample and test cable conditions shall be as specified in 1 and 2:
1. The test sample shall be tested on a ground plane. There shall be no insulated materials between the test setup and the ground plane.
2. The test cable running between the test sample and the damped sinusoidal generator shall be kept 4 cm (1.6 in.) to 6 cm (2.4 in.) above the ground plane.

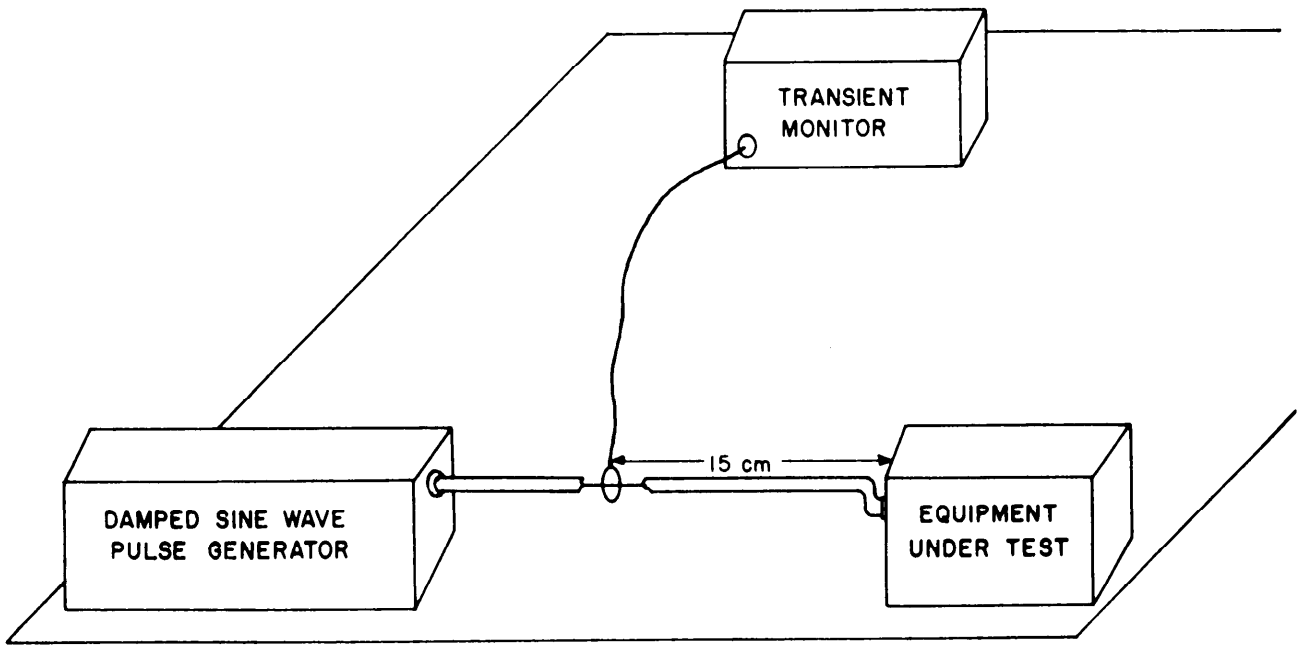


FIGURE 5. EMP test setup.

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4.5.10 Enclosure test. The IAU enclosure shall be subjected to the enclosure test of MIL-STD-108 for dripproof enclosures. The actual drip test shall be conducted.

4.5.11 Human engineering tests. The IAU shall be tested for human engineering characteristics in accordance with the Test and evaluation paragraph of MIL-H-46855 and approved human engineering test procedures. The steady-state noise shall be measured in accordance with MIL-STD-740.

4.5.12 Safety verification. Inspections, analyses, and demonstrations shall be conducted on two first article units to verify conformance with 3.6.7. If the contractor makes any changes to materials, processes, or the design after the units are accepted, the contracting officer may require an inspection on the unit produced under the new conditions. If a unit fails the safety inspection, the contractor shall immediately investigate the cause of failure and report to the procuring activity the results and actions taken to correct units which were manufactured under the same conditions. No unit produced under a set of conditions shall be accepted until a unit, produced under those conditions, has passed the safety inspection.

4.5.12.1 Leakage current test. Leakage current shall be measured on one IAU during first article inspection. Thereafter, leakage current shall be measured on one IAU per 500 successive IAUs produced (the first leakage current measurement shall be applicable to the first 500 IAUs). Leakage current shall be measured at 115 V, 60 Hz. The IAU test connection shall be as specified in 4.5.12.2 and measurement shall be as specified in 4.5.12.3.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE IAU DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.12.2 IAU test connection. After power removal, each IAU directly connected to an external power source and units deriving power from the IAU shall be placed on an insulated surface. All safety ground conductors between the IAU and units deriving power from the IAU shall be intact. The safety ground conductor between the IAU and the source power shall be opened during the test. OBSERVE WARNING STATEMENT. The IAU shall be connected as shown in FIGURE 6 for single-phase source power.

4.5.12.3 Measurement. Leakage current shall be measured on the IAU in its normal operating configuration. IAU controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor, when equal to 7.5 V, represents 5 mA of leakage current. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part to ground for every combination of switch positions available in FIGURE 6. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.6 Reliability testing. A sample of 12 first article IAUs shall be subjected to a fixed length test of 9250 hours. Each first article IAU shall be tested for approximately 770 hours. For production reliability testing, a sample of six to 12 IAUs shall be selected at random from each set of 500 IAUs produced and shall be subjected to the test. An accept decision shall result if not more than two chargeable failures, as defined in MIL-STD-781, occur, or when the procuring activity approves the corrective action plan for all chargeable failures that occur during test.

4.6.1 Testing conditions. Testing shall be conducted under the conditions specified in a through c:

a. Temperature cycling shall consist of approximately 4 hours at 50°C, and approximately 4 hours at 0°C, repeated throughout the test.

b. The IAU shall be operated continuously.

c. Fixed sine vibrational stress shall be applied at a level of 2.2 G ±10 percent, at a single nonresonant frequency between 20 Hz and 60 Hz for a 20-minute duration for each hour of operating time. The vibration shall be applied at a single axis, in the direction causing the greatest stress to the PCBs.

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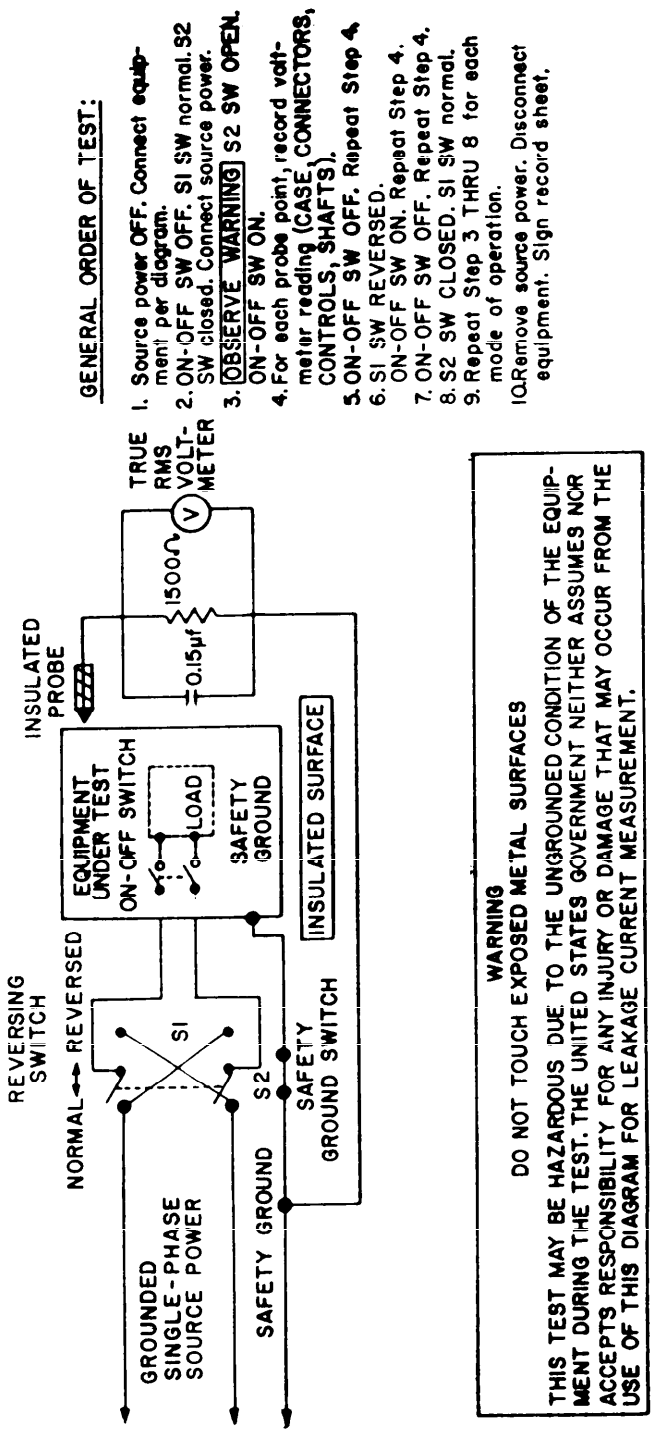


FIGURE 6. Single-phase test diagram for leakage current measurement.

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4.6.2 Performance tests. Performance tests shall be conducted on all IAUs to verify their normal operation during reliability testing. The performance tests shall be conducted at least once every normal 24-hour working day and shall, as a minimum, ensure the proper operation of the functions specified in a and b:

a. Observe operation of power-on indicator.
b. Observe operation of level converters. Ensure that operation remains as specified.

4.7 Maintainability demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be as specified in 4.7.1 through 4.7.5.

4.7.1 Organizational-intermediate maintainability. The contractor shall perform a maintainability demonstration in accordance with MIL-STD-471, Test Method 10. Only MIL-STD-471, Phase II, shall apply. The demonstration at the organizational level shall be performed by qualified technicians with experience equivalent to petty officer, third class. The demonstration shall be performed by replacement of subassemblies (individual PCBs on assemblies) and chassis-mounted parts. The maintainability demonstration shall be the means of demonstrating interchangeability. The demonstration shall be conducted in a space environment (including physical clearance) similar to that in which the IAU will be installed. Thirty candidate corrective maintenance tasks shall be identified in accordance with MIL-STD-471, APPENDIX A, and with the information specified in a through d:

a. Task number
b. Designation of specific faulty part
c. Failure mode
d. Means of introducing fault (for example, substitution of faulty part simulation, and so forth)

The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 20 tasks for the demonstration.

4.7.2 Accept or reject criteria. The demonstration shall be accepted for MTTR if eight trials are within the specified limits of a sample of 20; the demonstration shall be accepted for M_{maxct} only if all trials out of a sample of 20 are within the specified limits. After each maintenance task is completed, any degradation in system performance shall result in a reject decision.

4.7.3 Technicians. The procuring activity reserves the right to provide the technicians to perform the maintainability demonstration.

4.7.4 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the equipment.

4.7.5 Rejection. Failure to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies
b. Upon approval of an approach to the redesign or correction, the contractor shall implement the approach and repeat the demonstration until an accept decision is reached.

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4.8 Workmanship. Workmanship shall be as specified in 4.8.1 and 4.8.2.

4.8.1 General workmanship. The IAU, including subassemblies, shall be examined for workmanship and soldering during the fabrication and assembly process, and at the end item level, for conformance to 3.10.1. The workmanship examination shall include the examinations specified in 4.5.1a through 4.5.1r. Each solder connection and associated wiring or leads shall be visually examined.

4.8.2 Workmanship screen. Prior to conducting temperature cycling (see 4.8.2.2), vibration (see 4.8.2.1) shall be performed on each IAU. The vibration may be performed at the module, drawer, or end item level. In addition, temperature cycling and vibration shall be conducted on each spare power supply. All of the hardware, including cables and connectors, shall be exposed to vibration.

4.8.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All test IAUs shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 7. Input vibration levels shall be measured at the mounting points; the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz, and 100 Hz for vibration frequencies above 500 Hz, shall be used for the control and analyses of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level. The test IAU shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration restarted.

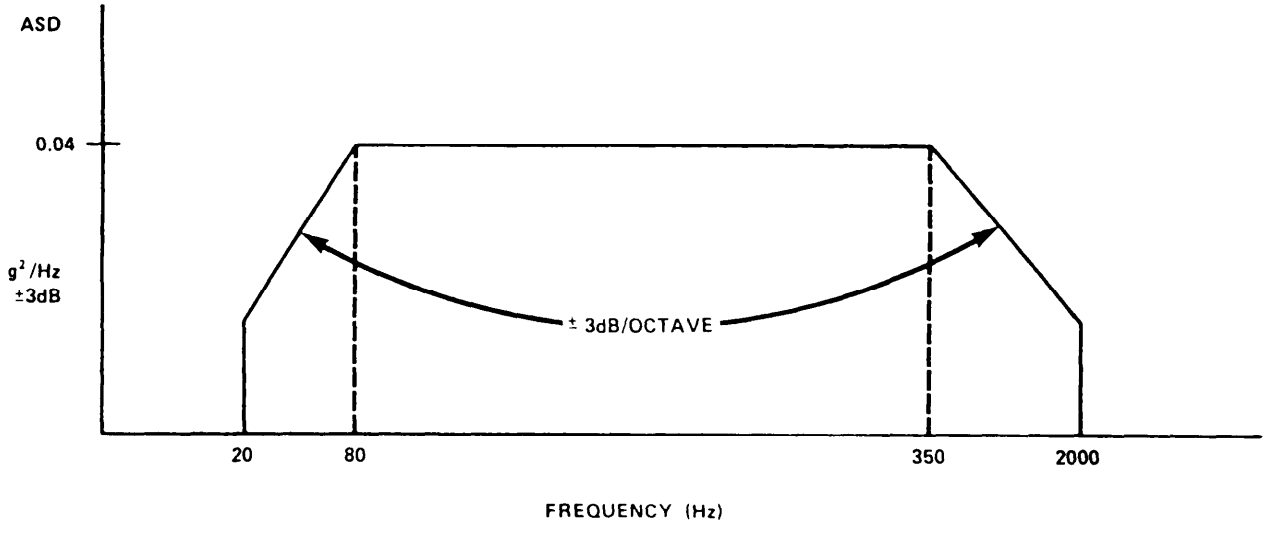


FIGURE 7. Random vibration curve.

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4.8.2.2 Temperature cycling. Each IAU shall be subjected to 10 cycles of the temperature curve shown in FIGURE 8. The temperature rate of change shall be not less than 5°C per minute. IAU power shall be turned on and off at the indicated times. The IAU shall be positioned for maximum exposure to the changing temperature. Where performance tests are called for in FIGURE 8, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the IAU reaches 80 percent of the chamber's temperature. When failures occur, the IAU shall be reworked and the cycling continued for a cumulative total of 10 cycles.

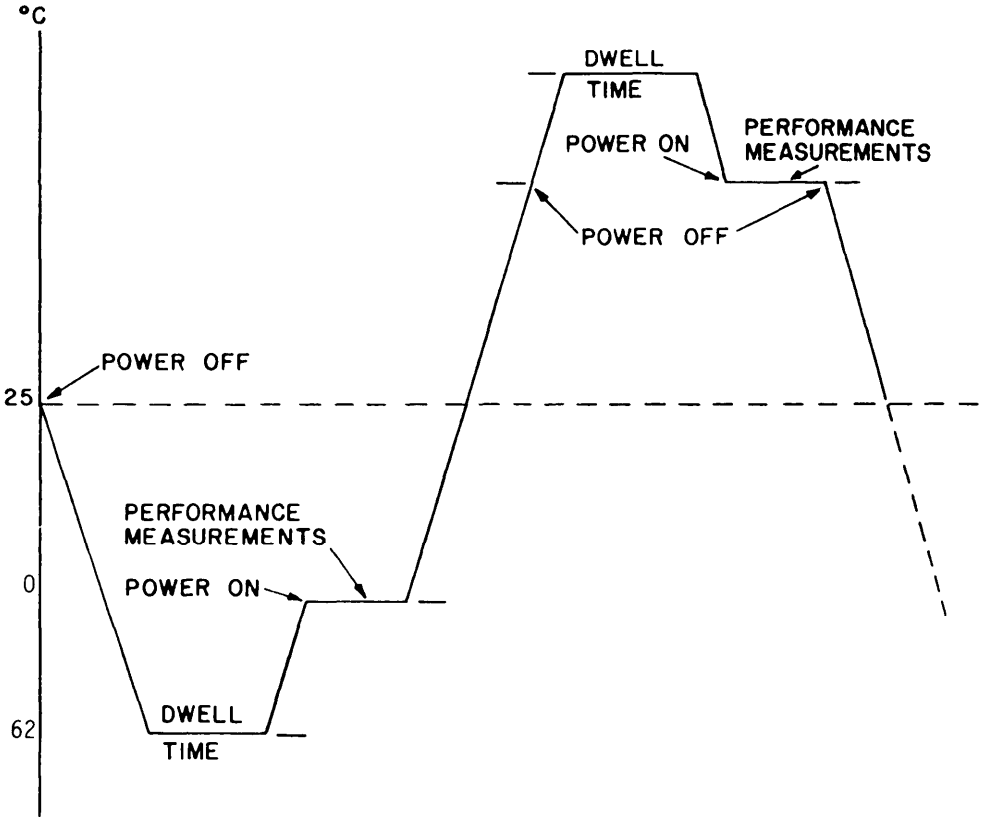


FIGURE 8. One cycle of temperature curve (not to scale).

4.9 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

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6. NOTES

6.1 Intended use. The IAU covered by this specification is intended for use with the KG-84 and secure shipboard communication systems which use TSEC/KW-7, TSEC/KW-26, TSEC/KG-13, and TSEC/KG-30 series cryptographic equipment.

6.2 Ordering data. Ordering data shall be as specified in 6.2.1 and 6.2.2.

6.2.1 Procurement requirements. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted (see 4.3)
c. Levels of preservation, packaging, packing, and marking (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
3.4.2	Environmental Design Criteria and Test Plan	DI-R-7125
3.6.1	TEMPEST Test Plan, Electromagnetics	DI-T-5140B
3.6.1	TEMPEST Test Evaluation Report, Electromagnetics	DI-T-5180A
3.6.1	TEMPEST Test Setup Ambient Signal Control Certification Report, Electromagnetics	DI-T-5183A
3.6.2, 3.6.3	Electromagnetic Interference Test Report	DI-R-7062
4.5.4	Environmental Test Report	DI-R-7127
4.5.8, 4.5.9	Electromagnetic Interference Test Plan	DI-R-7063

(DIDS related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of the number of IAUs specified in TABLE IX and 6.2.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-28892A(EC)	Interface Adapter Unit

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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